Year 8 Science
Biological Sciences
Investigation - Microscopes

INVESTIGATION 3.1

Getting into focus with an 'e'

AIM To practise focusing a monocular light microscope

Materials:

1 cm square piece of newsprint containing the letter 'e'
monocular light microscope
microscope slide
clear sticky tape
1 cm square piece of a coloured magazine or newspaper picture
hair strands (from different individuals)
spatula
selection of white powders and crystals (e.g. flour, salt, sugar, baking soda)
different brands or types of spices and leaf tea
fibres (e.g. cotton, linen, silk, wool, nylon)

METHOD AND RESULTS

- Carefully stick the 1 cm square of newsprint onto a clean microscope slide using sticky tape.
- Using the microscope directions, get the paper into focus using the coarse focus knob and the lowest power objective lens (smallest magnification).
- Carefully move the slide until you have a letter 'e' in focus.
- Change to a higher level of magnification by rotating to a higher power objective lens.

1. In which direction did the paper under the microscope move when you moved the slide (a) towards you or (b) to the left?
2. What does the letter 'e' look like under the microscope? Draw a pencil sketch of what you see.
3. Record the magnification that you use, and estimate how much of the viewed area is covered by the letter 'e' at this magnification.

- Using sticky tape, stick a selection of sample specimens onto microscope slides.

4. View your taped specimens using low power under the microscope and record your observations. Include detailed descriptions, the magnification used and an estimate of size next to your diagrams.

DISCUSS AND EXPLAIN

5. Suggest what the letters 'P' and 'R' would look like under the microscope. Sketch your predictions, and then view examples of these under the microscope. Were your predictions correct?
6. Summarise all your results in a table, using descriptive diagrams.
7. Summarise your microscopic observations of the sample specimens. Identify ways in which they were similar and ways in which they differed.
8. Identify details observed using the microscope that you could not see without the microscope.
9. On the basis of your experience, suggest advantages and disadvantages associated with light microscopes.
10. Propose a research question that you could explore using a light microscope, and describe how you could investigate it.